

PRANJAL KARAN

Software Engineering Intern Candidate · Data Structures, Algorithms & Full-Stack Development

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PROFESSIONAL SUMMARY

Computer Science undergraduate (CGPA 8.70/10, Expected May 2027) with strong fundamentals in **data structures, algorithms, and object-oriented programming** (C++, JavaScript, SQL), backed by a rigorous daily algorithmic problem-solving routine. Hands-on experience designing, building, and testing full-stack software – 3 end-to-end platforms shipped using **React.js, Next.js, Node.js**, and **WebSockets** – following solid engineering practices to improve database latency and deliver sub-100ms system performance. Quick to learn new tools (Azure DP-900 certified: 955/1000), collaborate across teams, and incorporate LLMs and IoT hardware to deliver quality solutions on tight timelines.

EDUCATION

VIT Bhopal University – Bachelor of Technology, Computer Science & Engineering | Sehore, MP

Sept 2023 – May 2027

CGPA: 8.70/10 | **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Web Development

TECHNICAL SKILLS

Languages	C++, JavaScript (ES6+), SQL, Python, Java, TypeScript
Frontend	React, Next.js, HTML5, CSS3, Tailwind CSS v4, Zustand, Vite, Responsive Web Design
Backend & APIs	Node.js, Express.js, RESTful API Design, WebSockets, JWT Authentication, Clerk Auth
Databases	MongoDB (Mongoose, Aggregation Pipelines), MySQL, PostgreSQL, Firebase, Firestore
Cloud & DevOps	Microsoft Azure (DP-900), Docker, Git, GitHub, Vercel, ImageKit
Tools & Practices	VS Code, Postman, OpenCV, ESP8266, Blynk IoT, Full-Stack Architecture, Agile / SDLC

PROFESSIONAL EXPERIENCE

ServiceNow (SmartBridge / AICTE) – Virtual Internship Program | Remote

May 2026 – Jun 2026

- **Applied** industry best practices to configure enterprise digital workflows and business rules utilizing ServiceNow Administration Fundamentals and FlowS.
- **Strengthened** IT operations testing and incident management reliability by leveraging Agentic AI concepts and the Automated Test Framework (ATF) to validate system updates.

PROJECTS

VibeChat – Real-Time Full-Stack Messaging Platform | [GitHub](#)

2026

React 19, Tailwind v4, Zustand, Node.js, MongoDB, Socket.io, Clerk, ImageKit, Docker

- **Engineered** a real-time messaging application achieving **sub-100ms** message distribution and interactive live presence indicators utilizing custom Socket.io server channels.
- **Accelerated** database query performance by replacing N+1 queries with a **6-stage Mongoose aggregation pipeline** and \$lookup joins, eliminating redundant roundtrips.
- **Built and tested** a lightweight **1.17MB** production bundle via Vite 8 featuring Clerk OAuth session gates, ImageKit lazy-loading pipelines, and a developer Mock Auth fallback.

HealConnect – Full-Stack Healthcare Monitoring Platform | [GitHub](#)

Jan 2026 – Present

Next.js, React, C++/Arduino, Firebase, Firestore, WebSockets, Blynk IoT, Gemini

- **Architected** a **13.1K-line** Next.js platform spanning **44 pages** and **23 reusable components**, integrating Firebase phone auth, EmailJS, and Gemini AI APIs.
- **Designed** C++/Arduino firmware for wearable ECG kits, achieving sub-second live data streaming via Blynk IoT, WebSockets, and Firestore onSnapshot listeners.
- **Delivered** role-based dashboards (Patient, Doctor, Admin) backed by **3 secure serverless APIs** and async polling loops for automated prescription processing.

LiveClass – Real-Time Video Learning Platform | [GitHub](#)

2026

MongoDB, Express.js, React, Node.js, JWT, REST APIs

- **Developed** a scalable MERN application managing the complete lifecycle of live video classroom sessions with persistent historical data logging in MongoDB.
- **Secured 100%** of authenticated REST API endpoints by implementing JWT authentication and Bcrypt password hashing across backend services.

ACHIEVEMENTS & CERTIFICATIONS

Achievements **National Qualifier** – Cleared Round 1 of **TCS CodeVita**, a national-level competitive coding challenge

Certifications **Microsoft Certified: Azure Data Fundamentals (DP-900)** (Score: **955/1000**) | **Applied Machine Learning in Python** (Coursera) | **AEO Fundamentals** (HubSpot) | **McKinsey Forward Program**

Leadership **Co-Lead (Social Media)**, Linpack Club, VIT Bhopal – Spearheaded campaigns driving a **30%** increase in event registrations and co-organized MatLabVerse.